# Introduction

House price prediction is an important application of machine learning in the real estate industry. Predicting house prices accurately helps buyers, sellers, investors, and real estate companies make informed decisions.

This project focuses on developing a machine learning model capable of predicting house prices based on important housing features such as overall quality, living area, garage size, and basement area. Different machine learning algorithms were trained and evaluated to determine the best-performing model.

# Problem Statement

Determining the correct price of a house can be difficult because house prices depend on multiple factors such as size, quality, garage capacity, and location-related features. Manual estimation may lead to inaccurate pricing.

The goal of this project is to develop a machine learning-based system that can predict house prices accurately using housing features from a real estate dataset.

# Objectives

- To analyze the house price dataset

- To clean and preprocess the dataset

- To identify important features affecting house prices

- To train machine learning models for prediction

- To evaluate model performance using different metrics

- To develop a house price prediction system

# Dataset Description

The dataset used for this project was obtained from the Kaggle House Prices dataset. The dataset contains multiple housing features such as overall quality, living area, garage area, basement size, and sale price.

The target variable for prediction is SalePrice.

# Exploratory Data Analysis


Exploratory Data Analysis was carried out to better understand the dataset before training the machine learning models.

The `df.shape` function was used to check the number of rows and columns in the dataset. This helped in understanding the size of the dataset and the number of available features.

The `df.columns` function was used to display all the column names in the dataset. This helped identify the available housing features and the target variable used for prediction.

The `df.info()` function was used to examine the dataset structure, including data types and missing values. This provided insight into which columns contained numerical or categorical data.

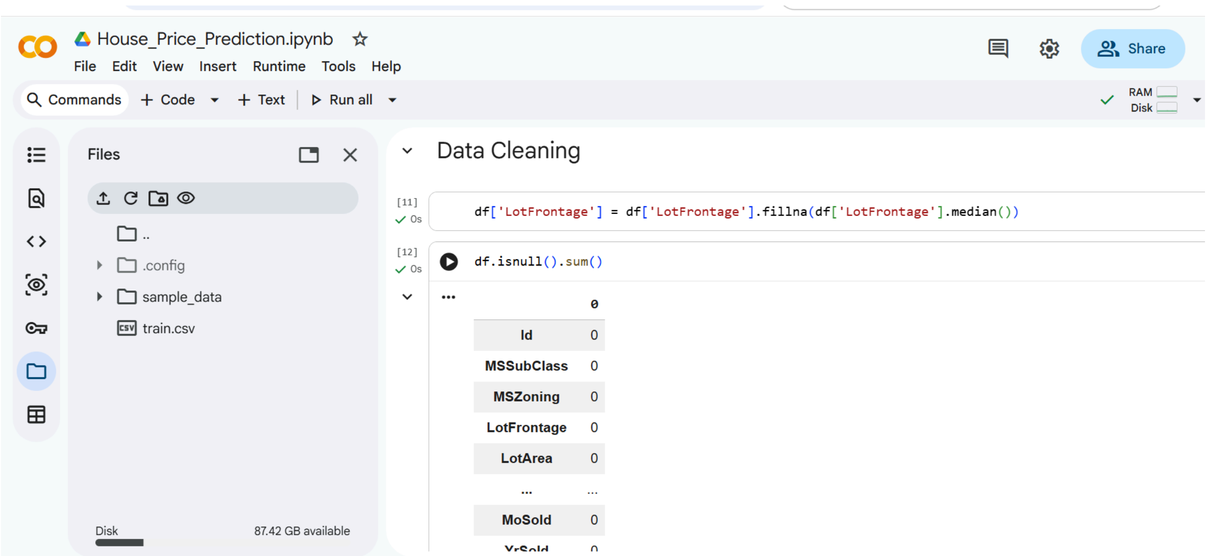
The `df.describe()` function was used to generate statistical summaries such as mean, minimum value, maximum value, and standard deviation for numerical columns in the dataset.

The `df.isnull().sum()` function was used to identify columns containing missing values. It was discovered that the `LotFrontage` column contained missing values.

Data visualization techniques were also applied during the analysis stage. Histograms were used to visualize the distribution of house prices, while scatter plots and heatmaps were used to analyze relationships between features and the target variable.

Correlation analysis was performed using the `corr()` function to identify features strongly related to house prices. Features such as `OverallQual`, `GrLivArea`, and `GarageCars` showed strong positive correlation with `SalePrice`.

## Data Cleaning




Data cleaning was performed to prepare the dataset for machine learning model training.

The `df.isnull().sum()` function was used to check for missing values in the dataset. It was discovered that the `LotFrontage` column contained 259 missing values, while the remaining columns had no missing data.

To handle the missing values, the median value of the `LotFrontage` column was used to replace the empty entries. The median method was chosen because it is less affected by extreme values and is suitable for real estate datasets.

The following code was used to fill the missing values:

```
df['LotFrontage'] =
df['LotFrontage'].fillna(df['LotFrontage'].median())
```

After filling the missing values, the dataset was checked again using `df.isnull().sum()` to confirm that all missing values had been successfully removed.

This preprocessing step ensured that the dataset was clean and ready for machine learning model training.

## Feature Selection

Feature selection was performed to choose the most relevant housing features for predicting house prices.

Correlation analysis was used to identify the features that had the strongest relationship with the target variable, `SalePrice`. Based on the correlation results, the following features were selected for model training:

- OverallQual
- GrLivArea
- GarageCars
- GarageArea
- TotalBsmtSF

These features showed strong positive correlation with house prices and were considered important predictors for the machine learning models.

The selected features were stored in the variable `X`, while the target variable `SalePrice` was stored in the variable `y`.

The following code was used for feature selection:

```
X = df[['OverallQual', 'GrLivArea', 'GarageCars',
'GarageArea', 'TotalBsmtSF']]

y = df['SalePrice']
```

This step helped reduce unnecessary features and improved the efficiency of the machine learning models.

## Model Training

The dataset was divided into training data and testing data using the `train_test_split()` function from Scikit-learn. The training data was used to train the machine learning models, while the testing data was used to evaluate model performance.

The dataset was split using 80% training data and 20% testing data with the following code:

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.2, random_state=42)
```

Three machine learning models were trained in this project:

- Linear Regression
- Decision Tree Regressor
- Random Forest Regressor

The Linear Regression model was trained using the `fit()` function to learn the relationship between housing features and house prices.

The Decision Tree Regressor model was trained to capture more complex relationships within the dataset by creating decision-based splits.

The Random Forest Regressor model combined multiple decision trees to improve prediction accuracy and reduce overfitting.

After training, predictions were made on the testing dataset using the `predict()` function for each model.

## Model Evaluation


The performance of the machine learning models was evaluated using different evaluation metrics, including Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and $R^2$ Score.

The evaluation metrics were used to measure the accuracy and prediction performance of each model.

The Linear Regression model achieved an $R^2$ score of approximately 0.79, indicating good prediction performance.

The Decision Tree Regressor achieved an $R^2$ score of approximately 0.81, performing slightly better than the Linear Regression model.

The Random Forest Regressor achieved the best performance with an $R^2$ score of approximately 0.88 and lower prediction errors compared to the other models.

The results showed that the Random Forest Regressor was the most accurate model for predicting house prices in this project.

The table below summarizes the model performances:

| **Model** | **$R^2$ Score** |
|---|---|
| Linear Regression | 0.79 |

| | |
|---|---|
| Decision Tree Regressor | 0.81 |
| Random Forest Regressor | 0.88 |

The Random Forest model was selected as the final model because it achieved the highest prediction accuracy and the lowest error values.

## Feature Importance

Feature Importance analysis was performed using the Random Forest Regressor model to determine which housing features contributed most to house price prediction.

The `feature_importances_` function was used to calculate the importance score of each selected feature. The results showed the following importance values:

| **Feature** | **Importance Score** |
|---|---|
| OverallQual | 0.586915 |
| GrLivArea | 0.208590 |
| TotalBsmtSF | 0.116505 |

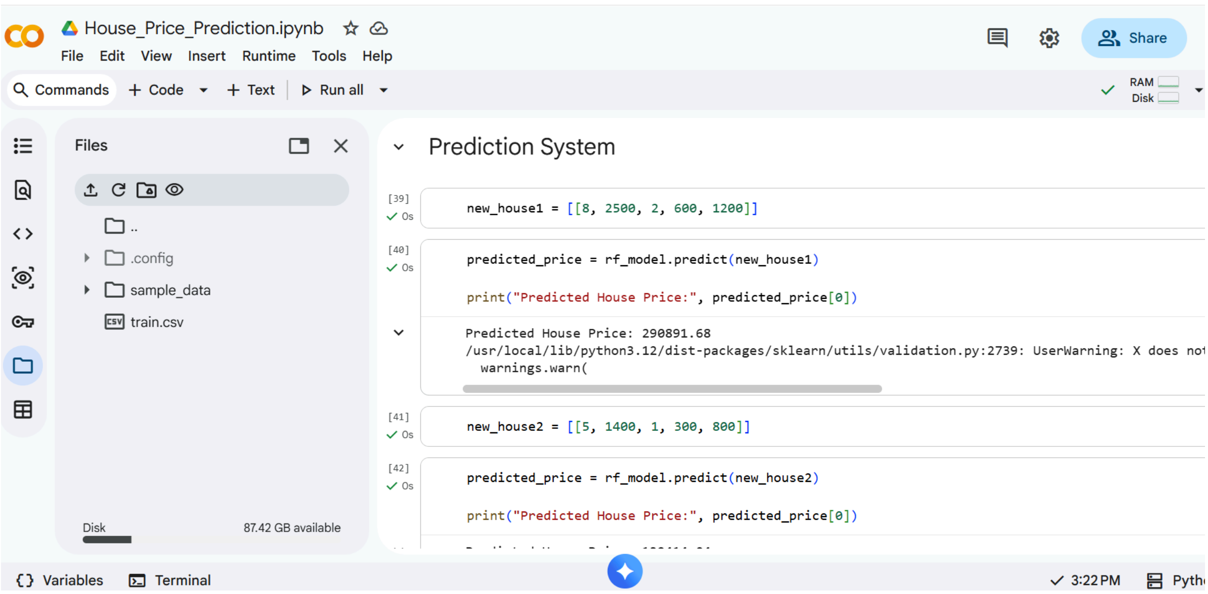
| | |
|---|---|
| GarageArea | 0.060801 |
| GarageCars | 0.027189 |

The analysis revealed that `OverallQual` had the highest influence on house price prediction, followed by `GrLivArea` and `TotalBsmtSF`.

This indicates that the overall quality of a house and the living area are the most significant factors affecting house prices in the dataset.

A bar chart was also created to visualize the feature importance values for better interpretation of the model results.

## Prediction System




After training and evaluating the machine learning models, the Random Forest Regressor was selected as the final prediction model because it achieved the best performance among all tested models.

The trained model was then used to predict house prices for new house samples using selected housing features such as overall quality, living area, garage capacity, garage area, and basement size.

Three different house samples were tested:

| **House Sample** | **Overall Qual** | **GrLivArea** | **GarageCars** | **GarageArea** | **TotalBsmtSF** | **PP** |
|---|---|---|---|---|---|---|
| new_house1 | 8 | 2500 | 2 | 600 | 1200 | 290891.68 |
| new_house2 | 5 | 1400 | 1 | 300 | 800 | 132414.04 |
| new_house3 | 10 | 4000 | 4 | 1000 | 2000 | 566582.76 |

The trained Random Forest model successfully generated predictions for the new house samples based on the provided housing features.

The prediction system demonstrates that the machine learning model can estimate house prices for unseen houses using important real estate features. This shows the practical application of machine learning in real estate price estimation and prediction systems.

# Conclusion

This project successfully developed a machine learning-based house price prediction system using housing data from the Kaggle House Prices dataset.

Exploratory Data Analysis (EDA) was performed to understand the dataset, analyze feature relationships, and identify important variables affecting house prices. Data cleaning techniques were applied to handle missing values and prepare the dataset for model training.

Three machine learning models were trained and evaluated in this project:

- Linear Regression
- Decision Tree Regressor

- Random Forest Regressor

Among the tested models, the Random Forest Regressor achieved the best performance with the highest $R^2$ score and the lowest prediction error values.

Feature Importance analysis showed that `OverallQual` and `GrLivArea` were the most influential features affecting house price prediction.

Finally, the trained Random Forest model was successfully used to predict prices for new unseen houses using selected housing features.

This project demonstrates the effectiveness of machine learning techniques in solving real-world real estate prediction problems.